Mini Project
Tube manipulator robot.

Team members

Last name	First name	Last name	First name	**Students**

Prepared by:

Name	Signature	Date	**One team member**

Controlled by:

Name	Signature	Date	**Second team member**

Approved by:

Name	Signature	Date	**Supervisor**

Degree Programme
Systems Engineering

Sion, Mai 13, 2026

User Requirements Specification

1. Time and Date

URS ID	Description
1.1	When referring to a *TimeStamp*, we mean a value like 2026-05-12 08:54:04 into a file, or 2026_05_12_08_54_04 if it needs to be used in a filename.
1.2	By default, Zurich time shall be used as time zone

2. Simplified Control

URS ID	Description
2.1	This project shall present for simplified machine control a single page accessible via an icon: test-tube in sidebar
2.2	A description page shall be integrated and commented in the SDS, Software Design Specification.
2.3	Simplified control with side-bar option fixed.
2.4	Simplified control with notebook layout.

3. Machine Startup

URS ID	Description
3.1	The machine shall start with a standard startup sequence: Clear, Reset, Start, which then transitions to the Stopped, Idle, and Execute states, respectively.
3.2	A Stop button shall allow switching from the Execute or Idle state to the Stopped state.
3.3	When the machine is in the Execute state, the X, Y, and Z axes shall be synchronized and ready for a Pick or Place sequence initiated by the operator.

4. Pick and Place Sequence

URS ID	Description
4.1	The operator shall be able to select a nest number between 1 and 13.
4.2	The operator shall be able to select a speed between 0.01 and 0.3 m/s and an acceleration between 0.1 and 3 m/s^2.
4.3	The operator shall be able to initiate either a Pick sequence or a Place sequence.

5. Log events

URS ID	Description
5.1	Each action on the simplified control page shall be logged in a **logTubeEvent.json** file
5.2	The action log shall contain the *timestamp* and the button name.
5.3	It's a plus to be able to include the username, saved in your Node-RED's User Settings in the json object.
5.4	It's a plus if buttons are built with customisable subflows.
5.5	Each alarm and warning shall be logged in separate file **logAlarmsAndWarning.json**.

6. Trace pick and place

URS ID	Description
6.1	Each robot movement shall be logged in a separate CSV file.
6.2	The file name shall be: **PickTimeStamp.csv** or **PlaceTimeStamp.csv**. see URS ID 1.
6.3	This file shall contain the on each row a timestamp;X; Y; and Z position records in [mm], as well as the Sound sensor dimension in [mm] and the ambient temperature in [°C]. Two decimal places are used.
6.4	The values shall be separated by semi-column

7. Backup parameters

URS ID	Description
7.1	It shall be possible to store an XML file at any time that contains the list of *PackTag.Command.Parameter_Lreal*, including the list of nest positions for the tubes.
7.2	The file name shall be: **CmdTimeStamp.xml** see URS ID 1.
7.3	The file shall contain the X, Y, and Z position records in [mm], as well as the Sound sensor dimension in [mm] and the ambient temperature in [°C]. Two decimal places are used.

8. Version

URS ID	Description
8.1	The software shall provide a list of all the versions needed to load the **flows.json** on a new machine.

9. Archiving

URS ID	Description
9.1	The URS, FS, DS, IQ, OQ, and PQ documents shall be submitted in hard copy.
9.2	A final report and your JSON file shall be archived on a Moodle repository for each group.